

When Does Information Sharing Improve Decentralized Discovery?

Aggregation, Independent Rescue, and Equilibrium Selection

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Information sharing changes both posterior quality and the diversity of actions available for discovery. We study that joint effect in finite one-hit search. First, exact channel comparisons show that one-person Bayes accuracy does not determine the value of a pooled action budget. Second, an incremental-sharing identity separates the accuracy gained by enlarging a pooled block from the independent rescue action that pooling removes. This yields an exact local criterion: sharing helps precisely when pooled residual error contracts faster than the removed private-failure factor. A centralized posterior top- L portfolio can recover the declared private baseline at full action capacity, but this is an authority benchmark, not a decentralized implementation.

We then study a two-agent equal-split discovery game with a hidden mixture of common and conditionally independent signal sources. At signal accuracy $p = 3/5$, sharing strictly improves discovery and each symmetric agent's payoff over the private anonymous symmetric Bayes–Nash selection exactly when $\rho \in ((5\sqrt{73} - 17)/48, 1)$. This theorem is selection-dependent: the shared outcome is the declared anonymous posterior-only, provenance-blind, identical-mixing equilibrium, not an every-equilibrium result. Opposite constant-target private equilibria exist and attain discovery one, while ownership-aware symmetric strategies can split targets on public disagreement. Sharing reveals the agreement pattern and changes posterior beliefs about dependence; it does not reveal the realized common-versus-independent source branch. The combined evidence is analytic and exact finite computation. It uses no human or real data, and the bounded channel registry is not a universal information order.

Keywords: information sharing; decentralized discovery; portfolio search; information aggregation; equilibrium selection; organizational design.

Evidence status. The manuscript synthesizes DD-C-0089 through DD-C-0110 without creating a new scientific claim. Analytic identities and theorems, independently reproduced bounded computations, a verified bounded negative result, and a verified equilibrium-

selection failure retain their ledger statuses. The source data are synthetic and model-generated.

Publication status. Working paper; not submitted, not peer reviewed, and no DOI.

1 Introduction

Should searchers share what they know before acting? The familiar answer is that sharing improves a group’s estimate of which action is best. That answer is incomplete when the outcome of interest is discovery by a portfolio of actions. Sharing can make one common action more accurate while removing independent attempts that could rescue a mistaken common choice. The organizational question is therefore not whether the pooled posterior is better. It is whether its gain is large enough to compensate for the search capacity absorbed by pooling.

This paper develops that comparison in four layers. First, an action-budget profile records the best centralized discovery probability obtainable from a pooled signal when the planner may take one, two, or more posterior-ranked actions. Second, an incremental-sharing identity decomposes a move from s to $s+1$ pooled agents into the improvement of the pooled block and the loss of one private rescue action. Third, a residual-error theorem gives the exact frontier: sharing helps precisely when the pooled error contracts faster than the rescue technology can repair it. Fourth, a binary Bayesian game shows why a positive selected-equilibrium result does not settle the decentralized implementation question.

The central nonstrategic identity is especially transparent. Let C_s be the probability that the correct candidate is covered by the action budget assigned to a pooled block of s agents. If each of the remaining $N - s$ independent searchers succeeds with probability q , the registered protocol yields

$$G_s = 1 - (1 - C_s)(1 - q)^{N-s}. \quad (1)$$

Writing $e_s = 1 - C_s$, an additional sharing step satisfies

$$G_{s+1} - G_s = (1 - q)^{N-s-1} [(1 - q)e_s - e_{s+1}]. \quad (2)$$

Thus the step helps if and only if $e_{s+1}/e_s < 1 - q$, is neutral at equality, and hurts above the threshold. The statement is exact under the stated rescue protocol (DD-C-0092 and DD-C-0097). It is not a theorem that more accurate signals always help, nor is it a comparison of arbitrary information structures.

The binary strategic result supplies a genuine positive sharing interval. Two agents receive individually accurate clues with $p = 3/5$ from a hidden mixture of a common-source branch and an independent-source branch. The realized branch is not revealed. Under the registered anonymous, label-equivariant private equilibrium and the posterior-only identical-mixing

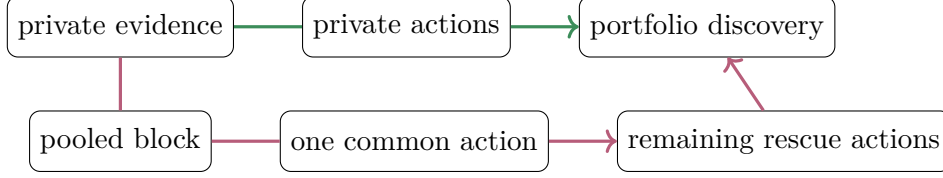


Figure 1: Information sharing changes two objects at once. The pooled block can improve one common action, while absorbing agents removes independent rescue actions. The comparison is therefore not a comparison of posterior accuracy alone.

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shared equilibrium, sharing strictly improves selected discovery exactly when

$$\rho \in \left(\frac{5\sqrt{73} - 17}{48}, 1 \right). \quad (3)$$

The same interval improves selected per-agent payoff (DD-C-0106 and DD-C-0107). This is a coordination-free positive result: no planner assigns targets and no binding recommendation is added when signals are shared.

The qualification is equally central. The positive interval is selection-dependent and is not an every-equilibrium result. For every dependence level, private play also has opposite constant-target pure equilibria that attain discovery one. After sharing, ownership-aware disagreement strategies can split targets without being functions only of the common posterior. Those alternatives show that posterior-only identical mixing is a selection, not the whole strategy space (DD-C-0109). A centralized top-two planner also obtains discovery one and strictly dominates the selected shared outcome (DD-C-0110). The positive theorem and the selection failure are presented together because separating them would overstate what information sharing implements.

Three further exact findings organize the analysis. One-person accuracy is not a sufficient statistic for portfolio value: two half-accurate channels can have the same direct private discovery yet opposite sharing directions and different centralized recovery budgets (DD-C-0089–DD-C-0091). Full action capacity always recovers at least the direct private baseline, but the minimum recovery budget depends on the entire signal geometry (DD-C-0098 and DD-C-0101). Finally, a complete 177-scenario registry contains compression, aggregation, and all-neutral curves but no mixed curve. The zero is preserved as a bounded negative result, not promoted to a general monotonicity theorem (DD-C-0099–DD-C-0103).

The paper contributes a theorem-family synthesis rather than a novelty claim. The canonical Shared Discovery Paradox provides motivating context and a conceptual precedent for the atomic information/action separation [Nakajima, 2026b]; it is not a foundational dependency for the sharing identity used here. That identity and its proof are self-contained in the incremental-sharing theorem and Appendix A and retain their DD-020 and DD-

021 authorities regardless of the companion paper’s status. The companion Incentive to Ignore paper owns selective attention and audience design [Nakajima, 2026a]. Decision theory studies comparisons of experiments [Blackwell, 1953]; team theory studies decentralized use of information [Radner, 1962]; social learning studies herding and cascades [Banerjee, 1992, Bikhchandani et al., 1992, Acemoglu et al., 2011]; and information design studies the consequences of signal structures and recommendations [Kamenica and Gentzkow, 2011, Bergemann and Morris, 2016]. Our narrower object is the composition of pooled posterior coverage, remaining rescue actions, and equilibrium selection in finite discovery portfolios. The literature search and evidence map distributed with the paper explain why broad priority language would be unsafe.

The rest of the paper proceeds from definitions to geometry, the sharing frontier, recovery, and strategic implementation. Every displayed numerical asset is generated from checksum-validated immutable run outputs. Exact numbers are labeled as fractions; decimals are displays. Centralized benchmarks, decentralized protocols, selected equilibria, and alternative equilibria are not interchanged.

2 Discovery architectures and comparison baselines

Let $\Theta = \{1, \dots, M\}$ contain exactly one correct candidate. Discovery occurs when at least one realized action equals the state. The objective is the ex ante probability of discovery. A channel maps the state to a signal. Agents may use signals privately, pool some signals, or submit them to a planner. The same evidence can therefore enter architectures with different action authority.

Definition 1 (Direct private baseline). With N private agents, each agent maps its own signal to one action under the registered direct rule. For a point signal (including a confidence-tagged point), the action is its nominated target. For a shortlist, the action is drawn uniformly from the reported set; for an exclusion signal, it is drawn uniformly from the non-excluded targets. Equivalently, set-valued signals use uniform randomization over the registered private Bayes-action correspondence, with independent private randomization across agents. The resulting discovery probability is P_N .

Definition 2 (Centralized action-budget profile). Given all signals in a pooled block, let

$$V_L = \mathbb{E} \left[\sum_{j=1}^L \pi_{(j)} \right],$$

where $\pi_{(1)} \geq \dots \geq \pi_{(M)}$ are the ordered posterior masses and $L \leq M$. The recovery budget is

$$L^* = \min\{L : V_L \geq P_N\}.$$

The definition separates information from capacity. V_1 measures the best single common action after pooling. V_L measures what a planner can cover with L distinct posterior-ranked

channel	signal geometry	q	P_3
noisy-point-half	symmetric-noisy-point	1/2	7/8
noisy-shortlist-three-quarters	noisy-k-shortlist	3/8	387/512
guaranteed-shortlist-two	guaranteed-shortlist	1/2	7/8
explicit-exclusion	exclusion	1/3	19/27
confidence-point	confidence-augmented-point	5/8	485/512

Table 1: Registered channel definitions and named comparison baselines. Accuracy and private discovery are exact; descriptions abbreviate the versioned channel records.

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actions. Neither number is the discovery of a decentralized equilibrium unless an institution implements those actions.

Four comparisons recur. The *private baseline* allows autonomous direct actions from private signals. The *pooled centralized profile* gives a planner the pooled posterior and an explicit action budget. The *incremental-sharing protocol* pools s agents into a common block while the others retain independent rescue actions. The *strategic protocol* lets agents maximize a shared discovery prize and requires an equilibrium rule. These are different counterfactuals, not alternative estimators of one quantity.

The authority distinction matters for interpretation. A planner that chooses the posterior top two can enforce diversification. Two autonomous agents who see the same posterior still face a coordination problem: absent roles, identifiers, or ownership-conditioned strategies, their symmetric response may duplicate. Conversely, two private agents can coordinate on opposite constant targets even without using their signals. That behavior is an equilibrium in the registered binary payoff game but is not direct clue-following. Comparing the wrong rows can reverse the verbal conclusion without changing any number.

The information structures themselves also differ. A point channel names one candidate; a shortlist identifies a set; an exclusion signal rules out a candidate; and a confidence point reports a candidate with a confidence class. Set-valued prediction is studied as a controlled ambiguity problem [Mortier et al., 2021, Sadinle et al., 2019]. Here those geometries are useful because they can hold one-person accuracy fixed while changing posterior rank mass and therefore multi-action coverage.

Boundary. The finite channels are registered synthetic objects. They are not estimated representations of laboratories, firms, committees, or human teams. The analysis compares discovery architectures within those objects.

3 Signal geometry is not one-person accuracy

The first result is a sufficiency failure. One-person accuracy q records the probability that a direct action is correct. It discards how posterior error is distributed across the remaining candidates. Yet a second or third action is valuable precisely because of that distribution.

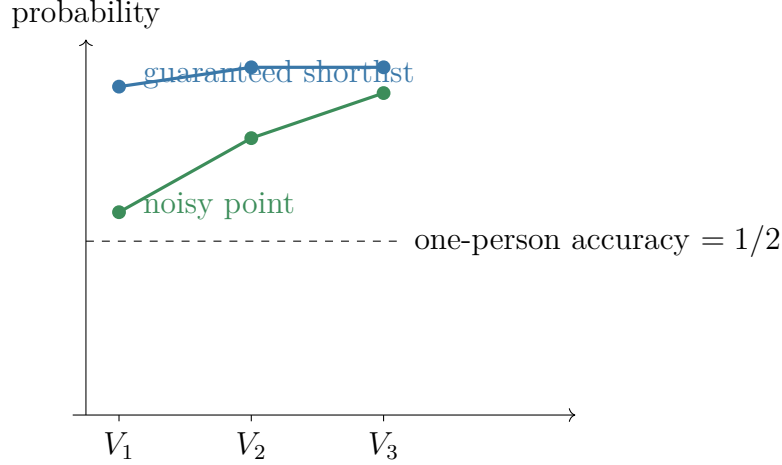


Figure 2: Same one-person accuracy, different pooled action-budget profiles. Both channels also have direct private discovery $7/8$, yet the noisy point requires three pooled actions to recover that baseline and the guaranteed shortlist requires one. Exact values are in table 2.

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channel	(V_1, V_2, V_3)	decimals	L^*
noisy-point-half	$(7/12, 43/54, 25/27)$	$(0.583, 0.796, 0.926)$	3
noisy-shortlist-three-quarters	$(35/64, 79/96, 539/576)$	$(0.547, 0.823, 0.936)$	2
guaranteed-shortlist-two	$(17/18, 1, 1)$	$(0.944, 1.000, 1.000)$	1
explicit-exclusion	$(16/27, 26/27, 1)$	$(0.593, 0.963, 1.000)$	2
confidence-point	$(295/384, 175/192, 31/32)$	$(0.768, 0.911, 0.969)$	3

Table 2: Exact DD-019 action-budget profiles. Decimals aid reading; exact fractions govern comparisons and recovery budgets.

Runs: 20260722T084145Z_DD-019_a77bb786_04a5e9f0c5; claims: DD-C-0089, DD-C-0091; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: a7da8a20aa24.

Proposition 3 (Action-budget profiles; DD-C-0089). *In the frozen $M = 4, N = 3$ registry, the five channel families have the exact profiles reported in table 2. In particular, the half-accurate noisy point and guaranteed-shortlist channels both have $q = 1/2$ and direct private discovery $P_3 = 7/8$, but their centralized profiles are respectively*

$$(V_1, V_2, V_3) = \left(\frac{7}{12}, \frac{43}{54}, \frac{25}{27} \right) \quad \text{and} \quad \left(\frac{17}{18}, 1, 1 \right).$$

The computations are finite exact enumerations independently reproduced from the registered model and fixtures. The proposition does not rank all point and shortlist channels. It exhibits two channels for which equal accuracy and equal direct private discovery fail to determine pooled portfolio value.

Corollary 4 (Accuracy is insufficient; DD-C-0090). *Neither one-person accuracy nor the pair (q, P_3) determines the centralized action-budget profile or the minimum recovery budget.*

For the noisy point, one pooled action covers only 7/12 and two cover 43/54; three are required to recover 7/8. For the guaranteed shortlist, one pooled action already covers 17/18, so $L^* = 1$. The exclusion and confidence-point channels provide further distinct shapes: the former reaches certainty at three actions, while the latter has high direct private discovery 485/512 but needs three pooled actions to recover it.

Proposition 5 (Registered recovery budgets; DD-C-0091). *For noisy point, noisy shortlist, guaranteed shortlist, exclusion, and confidence point, the recovery budgets are respectively 3, 2, 1, 2, 3.*

The result suggests a practical diagnostic. An organization considering pooled evaluation should estimate or stress-test a coverage curve, not report only the top-choice accuracy. If pooled evidence concentrates almost all posterior mass on a small set, a modest common action budget can be enough. If its errors remain diffuse or systematically compress plausible alternatives behind one label, pooling may require many actions to restore the private portfolio baseline.

This diagnostic is related to, but not supplied by, Blackwell’s comparison of experiments. Blackwell dominance ranks experiments for every decision problem with a single decision maker [Blackwell, 1953]. The present comparisons hold the discovery problem fixed while changing how many actions remain and who controls them. A scalar accuracy ordering is weaker still. We therefore do not describe the bounded registry as a universal information order.

4 Aggregation gain and independent rescue

Consider N searchers. A pooled block of size s produces coverage C_s from its assigned action budget. Each of the $N - s$ remaining private searchers has an independent probability q of rescuing a miss. Conditional on a pooled miss, all private rescue actions fail with probability $(1 - q)^{N-s}$. This yields equation (1).

Theorem 6 (Incremental-sharing identity; DD-C-0092). *For $1 \leq s < N$,*

$$G_{s+1} - G_s = (1 - q)^{N-s-1} \{(1 - q)(1 - C_s) - (1 - C_{s+1})\}.$$

Consequently the sign of an additional sharing step is the sign of the pooled coverage gain net of the private rescue action it replaces.

Proof. Subtract $1 - (1 - C_s)(1 - q)^{N-s}$ from $1 - (1 - C_{s+1})(1 - q)^{N-s-1}$ and factor the positive term $(1 - q)^{N-s-1}$. No distributional approximation is used. $\square$

Corollary 7 (Noisy point monotonicity; DD-C-0093). *For the registered noisy point family, the coverage improvement from adding a signal does not overcome the lost rescue action; the incremental-sharing curve is weakly decreasing on the proved parameter domain.*

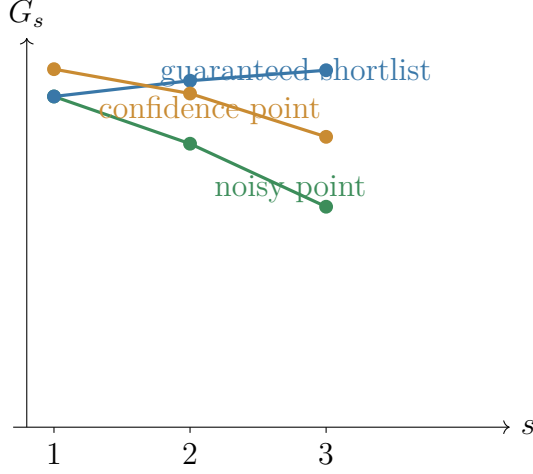


Figure 3: Incremental-sharing curves at $M = 4, N = 3$. The half-accurate noisy point declines, the equally accurate guaranteed shortlist rises, and the confidence-point channel declines. The sign is a property of channel geometry and protocol, not accuracy alone.

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channel	C_2	(G_1, G_2, G_3)	adjacent changes	sign
noisy-point-half	1/2	(7/8, 3/4, 7/12)	(-1/8, -1/6)	decreasing
noisy-shortlist-three-quarters	1/2	(387/512, 11/16, 35/64)	(-35/512, -9/64)	decreasing
guaranteed-shortlist-two	5/6	(7/8, 11/12, 17/18)	(1/24, 1/36)	increasing
explicit-exclusion	4/9	(19/27, 17/27, 16/27)	(-2/27, -1/27)	decreasing
confidence-point	11/16	(485/512, 113/128, 295/384)	(-33/512, -11/96)	decreasing

Table 3: Exact DD-020 sharing paths and intermediate two-signal pooled coverage. The pooled block grows from $s = 1$ to 3 while the remaining private rescue actions disappear.

Runs: 20260722T142551Z_DD-020_3854fff6_37c11a850a; claims: DD-C-0092, DD-C-0096; generator:

distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: a4f63d52f402.

The corollary is channel-specific. It does not follow from $q = 1/2$. The registered guaranteed-shortlist channel has the same accuracy and moves in the opposite direction.

Proposition 8 (Opposite registered paths; DD-C-0094). *At $M = 4, N = 3, q = 1/2$, the noisy-point path is $7/8, 3/4, 7/12$, with changes $-1/8, -1/6$, whereas the guaranteed-shortlist path is $7/8, 11/12, 17/18$, with changes $1/24, 1/36$.*

The DD-020 census has 2,555 protocol rows arranged into 511 parameter chains, each containing one row for every sharing-block size. Each chain contributes one initial row, so 2,555 rows minus 511 chain starts yields 2,044 adjacent transitions. The chain starts are not unclassified observations: the adjacent transitions are exactly 1,848 negative, 196 neutral, and zero positive (DD-C-0095). The registered grid is $M = 2, \dots, 8, N = 2, \dots, 8$, every declared accuracy in the 73 (M, p) cells, and all $s = 1, \dots, N$. Those counts reproduce the theorem’s sign on a bounded grid; they are not the proof. The guaranteed-shortlist row is a

verified counterchannel to any claim that sharing must decrease simply because the private portfolio loses independent actions (DD-C-0096).

The decomposition clarifies several verbal confusions. Sharing does not merely “add information.” It expands the pooled information set while changing the portfolio’s action structure. Nor does diversification merely “waste” the pooled posterior: a low-ranked private action can be valuable precisely on the event that the pooled block is wrong. The sign is controlled by the product of these two margins.

5 The General Sharing Frontier

The identity becomes a general criterion after expressing coverage in residual errors. Let $e_s = 1 - C_s$. When $e_s > 0$, define the contraction ratio $\rho_s = e_{s+1}/e_s$. The private rescue threshold is $1 - q$.

Theorem 9 (General Sharing Frontier; DD-C-0097). *Under the registered independent-rescue protocol with $q < 1$,*

$$\text{sign}(G_{s+1} - G_s) = \text{sign}((1 - q)e_s - e_{s+1}).$$

If $e_s > 0$, the step strictly improves discovery exactly when $\rho_s < 1 - q$, is neutral exactly when $\rho_s = 1 - q$, and strictly reduces discovery exactly when $\rho_s > 1 - q$.

Proof. The first statement is equation (2); its prefactor is positive for $q < 1$. Dividing the bracket by $e_s > 0$ gives the ratio criterion. If $e_s = 0$, pooled coverage is already one and the undivided expression governs the boundary. $\square$

The theorem is a break-even rule. The new pooled signal must shrink residual error by more than an independent private attempt would. A larger q makes rescue more powerful and lowers $1 - q$, so pooling faces a stricter contraction test. A weaker rescue technology raises the threshold and makes pooling easier to justify. The result is deliberately architectural: changing dependence among rescue actions or allowing the pooled block to control a different number of actions changes the formula.

The frontier reconciles apparently contradictory examples. The noisy point compresses error too slowly relative to a half-accurate rescue action. The guaranteed shortlist eliminates enough residual mass to compensate for losing that action. Equal q therefore places both examples against the same horizontal threshold while their error-contraction curves fall on opposite sides.

Proposition 10 (Bounded frontier classification; DD-C-0099 and DD-C-0100). *The frozen 177-scenario registry contains 126 strict-compression-dominated curves, 16 strict-aggregation-dominated curves, and 35 all-neutral curves. The stored error ratios reproduce the theorem’s sign in every registered row.*

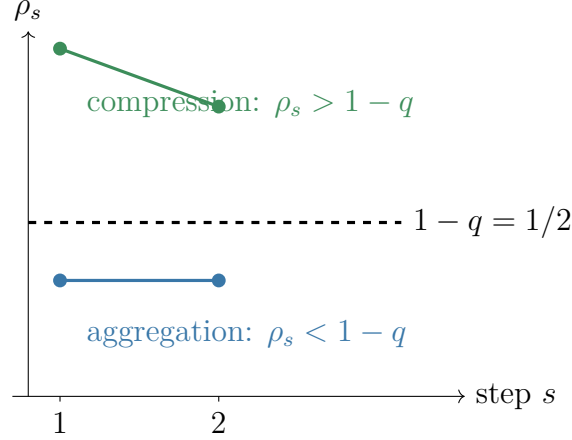


Figure 4: Residual-error frontier for the registered half-accurate $M = 4, N = 3$ point and guaranteed-shortlist channels. A sharing step helps below the rescue threshold, is neutral on it, and hurts above it.

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distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 3521286a3415.

The names describe the frozen curves. “Compression dominated” means residual error did not contract fast enough relative to rescue at each applicable step; “aggregation dominated” means it contracted faster; “all neutral” means the frontier held with equality. The registry is a systematic stress test across named finite channel families and parameter values, not a probability sample of real organizations.

For the full-sharing endpoint, the pooled block has $s = N$ and its one-action consensus coverage is C_N . The full-sharing labels in table 4 are generated directly from that endpoint: class B means $q < C_N < P_N$ (a shared-discovery paradox), class C means $C_N > P_N$ (strict aggregation-dominated consensus), and class D is the remaining equality boundary after those strict cases (so $C_N = q$ or $C_N = P_N$). In the frozen registry, B comprises 78 of the 126 strict-compression-dominated curves; C comprises all 16 strict-aggregation-dominated curves; and D comprises the 35 all-neutral curves plus the remaining 48 strict-compression-dominated curves. These are registry cross-tabs, not additional frontier theorems.

DD-021 and DD-020 deliberately use different scopes. DD-021 enumerates five registered channel families over $M = 3, 4, 5$, $N = 2, 3, 4$, and $L = 1, \dots, \min(N, M)$: 59 channel laws produce 177 scenarios for the general frontier and centralized recovery classification. DD-020 instead isolates the symmetric noisy-point monotonicity theorem on the broader $M = 2, \dots, 8$, $N = 2, \dots, 8$ declared accuracy grid, while carrying the five fixed DD-019 channels only as named comparison paths. The larger DD-020 row count tests one channel theorem across more parameters; the smaller DD-021 registry compares more channel geometries and recovery budgets on a deliberately bounded grid.

Proposition 11 (Minimal registered witnesses; DD-C-0100). *Under the frozen lexicographic ordering, the registry contains minimal witnesses for equal-baseline opposite signs, equal-*

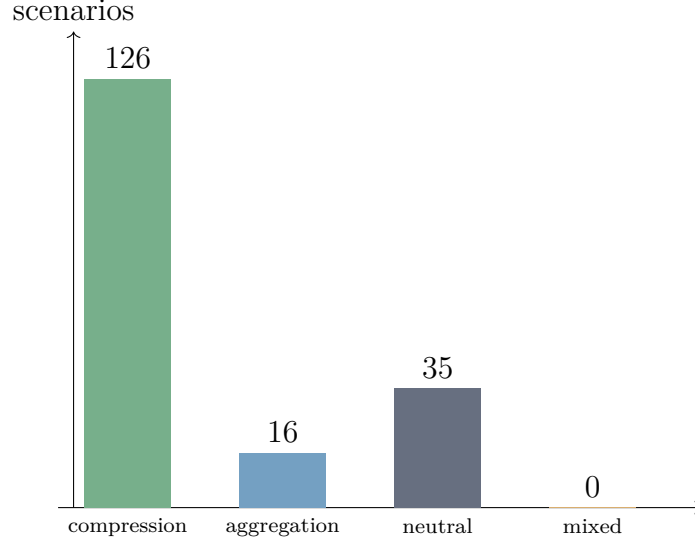


Figure 5: Complete bounded classification of 177 registered scenarios: 126 strict compression curves, 16 strict aggregation curves, 35 all-neutral curves, and no mixed curve. The zero mixed count is a bounded null, not a general theorem.

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distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 2bd510abf8cd.

accuracy recovery differences, a shared-discovery-paradox configuration, and consensus dominance. Their exact values appear in table 5.

Proposition 12 (Preserved bounded null; DD-C-0103). *No registered curve is mixed: none crosses from aggregation dominated to compression dominated, or conversely, over its recorded sharing steps.*

Boundary. The zero mixed count is a verified bounded negative result. It does not imply that arbitrary channels have monotone sharing curves. A mixed curve outside the registry would not contradict the theorem; it would simply place different steps on different sides of the exact frontier. The bounded zero also does not establish whether mixed curves are feasible within each registered family’s unrestricted parameterization; that structural question was not tested.

This distinction illustrates the paper’s evidence discipline. The analytic criterion applies wherever its assumptions hold. The class counts apply only to the frozen registry. The absence of a class is retained because negative results inform the next search, but it is not converted into a universal statement.

axis	class	count	status
sharing curve	all-neutral	35	exact bounded
sharing curve	strict-aggregation-dominated	16	exact bounded
sharing curve	strict-compression-dominated	126	exact bounded
full sharing	B-shared-discovery-paradox	78	exact bounded
full sharing	C-strict-aggregation-dominated-consensus	16	exact bounded
full sharing	D-boundary	83	exact bounded
recovery budget	$L^* = 1$	51	centralized
recovery budget	$L^* = 2$	55	centralized
recovery budget	$L^* = 3$	64	centralized
recovery budget	$L^* = 4$	7	centralized
sharing curve	mixed	0	bounded null

Table 4: Complete bounded DD-021 classifications and recovery budgets. Counts describe the frozen registry, not a population of channels or organizations.

Runs: 20260722T185924Z_DD-021_3cdbbc40_2fea269a9a; claims: DD-C-0099, DD-C-0101, DD-C-0102, DD-C-0103; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 2bd510abf8cd.

witness	left or focal value	right or comparator	conclusion
same baseline, opposite signs	point: $-1/4$	guaranteed shortlist: $1/12$	$M = 4, N = 2, q = 1/2$
same accuracy, recovery	point: $L^* = 2$	guaranteed shortlist: $L^* = 1$	$M = 3, N = 2, q = 1/2$
Shared Discovery Paradox	$q = 3/8$	$C_N = 27/64 < P_N = 39/64$	noisy shortlist
consensus dominance	$P_N = 3/4$	$C_N = 5/6$	guaranteed shortlist

Table 5: Minimal registered witnesses under the frozen lexicographic order. Minimality is internal to the registry.

Runs: 20260722T185924Z_DD-021_3cdbbc40_2fea269a9a; claims: DD-C-0100; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: e897c62593b4.

6 Centralized action-budget recovery

Pooling need not be evaluated at one common action. If a planner can assign several actions after observing pooled evidence, the top- L posterior coverage V_L is nondecreasing in L . At full candidate capacity it is one. The direct private portfolio cannot exceed one, giving a general recovery statement.

Theorem 13 (Full-capacity recovery; DD-C-0098). *For every registered finite channel with M candidates,*

$$V_M = 1 \geq P_N.$$

Hence a finite minimum recovery budget $L^ \leq M$ exists. Moreover, $L^* = 1$ if and only if $C_N = V_1 \geq P_N$.*

Proof. The posterior masses sum to one, so choosing all M candidates covers the state with probability one. Monotonicity of ordered partial sums implies existence of the first L reaching P_N . The last statement is the definition at $L = 1$. $\square$

The theorem is an existence result under centralized authority, not a claim that M decentralized agents will choose distinct actions. It also does not say that pooling is costless:

an organization may lack M actions or may value actions for other tasks. It identifies the precise capacity question that must be answered after pooling.

Proposition 14 (Bounded recovery census; DD-C-0101 and DD-C-0102). *In the frozen 177-scenario registry, recovery budgets $L^* = 1, 2, 3, 4$ occur in 51, 55, 64, 7 scenarios respectively. Every registered scenario satisfies the full-capacity theorem and the exact stored recovery characterization.*

The distribution demonstrates why a binary recommendation—share or do not share—is usually underspecified. For 51 scenarios, one centralized action after pooling already matches the direct private baseline. For 126, at least two actions are required, and seven need all four. The relevant design variable is the pair consisting of information architecture and post-sharing action budget.

The distinction also helps compare centralization with communication. A team can share signals without granting a planner binding authority; it can grant a planner an action budget without pooling all raw signals; and it can retain private rescue capacity after producing a common recommendation. These designs occupy different points in the space summarized by figure 1.

7 Coordination-free positive sharing

We now turn from nonstrategic rescue to a Bayesian game. There are two agents and two target labels. The state is equally likely. Each clue is correct with probability $p > 1/2$. A hidden branch is common with probability ρ and independent with probability $1 - \rho$. In the common branch the clue-generation shock is shared; in the independent branch it is not. Agents do not observe the realized branch. After actions, discovery pays one if at least one chosen label is correct, split equally among agents choosing that correct label.

Let $t = 2p - 1$ and

$$A = t^2 + \rho(1 - t^2).$$

Here A is the unconditional clue-alignment statistic: t^2 is its accuracy-driven component and $\rho(1 - t^2)$ is its source-dependence component. The source-dependence mixture probability ρ is distinct from the residual-error contraction ratio ρ_s in the General Sharing Frontier. The private selection is the registered anonymous, label-equivariant Bayesian Nash equilibrium. The shared selection uses identical mixing as a function of the common posterior only. These restrictions make the compared objects exact and reproducible, but they are selection assumptions.

Theorem 15 (Selected equilibrium formulas; DD-C-0104 and DD-C-0105). *In the private selection, the probability r of following one’s clue is*

$$r^* = \begin{cases} 1, & A \leq 3t, \\ \frac{1}{2} + \frac{3t}{2A}, & A > 3t. \end{cases}$$

information	state statistic	selected rule	scope
private	$t = 2p - 1, A = t^2 + \rho(1 - t^2)$	$r^* = 1$ if $A \leq 3t$; else $1/2 + 3t/(2A)$	anonymous label-equivariant BNE
shared	$u = 1/2 + t/(1 + A)$	$x^* = 1$ if $u \geq 2/3$; else $3u - 1$	posterior-only identical mixing
agreement			
shared	posterior $1/2$	$x^* = 1/2$	ownership-blind selection
disagreement			

Table 6: Exact DD-022 selected-equilibrium formulas. The selections are deliberately narrower than the complete pure correspondence.

Runs: 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims: DD-C-0104, DD-C-0105; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 98fb7d475365.

After sharing, agreement produces posterior correctness

$$u = \frac{1}{2} + \frac{t}{1 + A}.$$

The selected probability x of following the agreed label is

$$x^* = \begin{cases} 1, & u \geq 2/3, \\ 3u - 1, & u < 2/3, \end{cases}$$

while disagreement uses the symmetric probability $1/2$.

The formulas follow from indifference and boundary conditions in the two-action game. They do not assume that sharing reveals which dependence branch generated the clues. Agreement is informative about the state and about the likelihood of common generation; it does not disclose the realized common/independent branch.

Theorem 16 (Strict selected sharing gain at $p = 3/5$; DD-C-0106). *Let $p = 3/5$. Shared selected discovery is strictly greater than private selected discovery exactly for*

$$\rho \in (\rho^*, 1), \quad \rho^* = \frac{5\sqrt{73} - 17}{48}.$$

Equality holds at ρ^ and at 1. The certified isolating interval is $2679/5000 < \rho^* < 67/125$.*

Proof sketch. Substitute $t = 1/5$ into the regime formulas. The shared rule changes regime at $\rho = 1/6$ and the private rule at $\rho = 7/12$. On each interval the discovery difference is rational in ρ . Clearing positive denominators yields the certified algebraic equality with its unique interior root $(5\sqrt{73} - 17)/48$. Exact sign checks on the isolated intervals give a strict gain between that root and one, with equality restored at perfect dependence. The immutable certificate supplies the symbolic factors and rational isolating interval; no floating-point root is used in the theorem. $\square$

Corollary 17 (Selected payoff gain; DD-C-0107). *At $p = 3/5$, the same open interval strictly improves expected payoff per agent under the registered selections.*

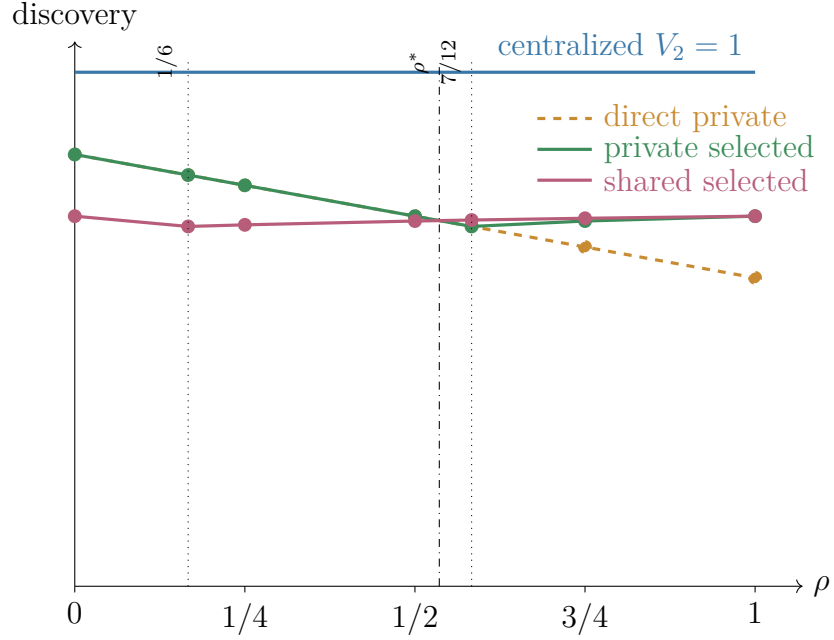


Figure 6: Discovery versus source dependence at $p = 3/5$. The exact theorem compares the private and shared selected equilibria, not direct clue-following. Selected sharing is strictly higher only on $(\rho^*, 1)$, where $\rho^* = (5\sqrt{73} - 17)/48$; equality returns at one. The selected shared outcome remains below centralized V_2 . Lines connect registered exact cells for display; the interval theorem comes from the analytic certificate.

Runs: 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims: DD-C-0104, DD-C-0105, DD-C-0106, DD-C-0110; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 98fb7d475365.

object	exact value	display	interpretation
shared regime boundary	$1/6$	0.1667	agreement rule changes
positive crossover ρ^*	$\frac{5\sqrt{73}-17}{48}$	0.5358337235	equality
private regime boundary	$7/12$	0.5833	private anti-crowding begins
planner gap, low regime	$(7 + 3\rho)/25$	positive	centralized $V_2 = 1$
planner gap, high regime	$(4 + 3\rho)/(13 + 12\rho)$	positive	centralized $V_2 = 1$

Table 7: Exact canonical threshold, regime boundaries, and centralized implementation gap at $p = 3/5$.

Runs: 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims: DD-C-0106, DD-C-0110; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 146dcfba3f74, 8ae5af15a2f7.

accuracy p	positive	neutral	negative
$1/2$	0	7	0
$11/20$	4	1	2
$3/5$	2	1	4
$13/20$	0	1	6
$2/3$	0	1	6
1	0	7	0

Table 8: Exact DD-022 selected-sharing gain classes by signal accuracy. Each row contains the seven registered dependence values; counts are bounded cells, not empirical frequencies.

Runs: 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims: DD-C-0108; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 98fb7d475365.

The mechanism is not “correlation is good.” At low dependence, private clues provide useful diversity and sharing induces avoidable crowding. On $(\rho^*, 7/12)$, private selected play still follows each clue fully; the gain comes from sufficiently informative agreement posteriors under the selected shared rule. On $(7/12, 1)$, the private selected response also becomes anti-crowding, reinforcing the selected sharing gain. At perfect dependence, sharing adds no information relative to identical private clues, and the selected values meet again. Dependence affects both posterior aggregation and strategic duplication, but anti-crowding is not the mechanism throughout the whole positive interval.

The bounded 7×6 registry contains 42 exact (p, ρ) cells: six positive, 18 negative, and 18 neutral comparisons (DD-C-0108). This is a complete count of the registered grid, not a measure of the interval’s size and not evidence about empirical frequencies.

The exact positive-interval theorem is intentionally fixed at $p = 3/5$. On the separate registered grid in table 8, positive cells appear at $p = 11/20$ and $p = 3/5$ and have count zero at the higher listed accuracies $p = 13/20$ and $p = 2/3$. This finite cross-tab describes the registered cells only: it neither proves a monotone accuracy mechanism nor rules out a positive interval outside the registered grid.

Private alternative	Shared selection	Shared broader strategy	Centralized benchmark
opposite constant targets	posterior-only identical mixing	ownership-aware disagreement split	binding top-two
⇓	⇓	⇓	⇓
discovery one	selected positive interval; planner gap	disagreement coverage beyond posterior-only rules	$V_2 = 1$ by authority

Figure 7: Equilibrium and authority map. The posterior-only identical-mixing outcome is one selected decentralized equilibrium, not the correspondence. Constant-target private equilibria and ownership-aware disagreement strategies expose the selection boundary; centralized top-two uses binding authority.

Runs: 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims: DD-C-0106, DD-C-0109, DD-C-0110; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: 98fb7d475365.

8 Equilibrium selection and implementation failure

The positive theorem compares two named equilibria. It does not state that sharing improves every equilibrium or even the best or worst equilibrium in the full correspondence. This qualification is mathematical, not rhetorical.

Theorem 18 (Selection failure; DD-C-0109). *For every ρ in the registered binary model, private play admits pure equilibria in which the two agents choose opposite constant targets. These equilibria attain discovery one. After sharing, ownership-aware disagreement strategies can assign different targets when clues disagree and lie outside the posterior-only identical-mixing selection. Therefore the strict interval in equation (3) is not an every-equilibrium improvement result.*

Argument. With opposite constants, exactly one agent selects the true binary target in every state. A unilateral deviation can only duplicate the other target in one state or exchange which state is covered, so it does not yield a strict payoff gain under the registered split-prize game. For shared disagreement histories, the agents’ ownership of the two private clues provides a role variable. Mapping those roles to opposite actions covers both targets, but the mapping is not a function solely of the common posterior. These constructions are checked symbolically in the DD-022 proof record. $\square$

Equilibrium selection is a substantive institutional primitive. Anonymous label-equivariant restrictions rule out fixed label roles. Posterior-only shared strategies rule out conditioning on who observed which clue. Identical mixing rules out asymmetric role assignment. Each restriction may be appropriate for an application, but none follows merely from

“the agents share information.” Classical equilibrium-selection theory similarly emphasizes that a game can admit many self-consistent predictions [Harsanyi and Selten, 1988].

Theorem 19 (Centralized implementation gap; DD-C-0110). *At $p = 3/5$, a centralized top-two planner attains $V_2 = 1$. Its gap above the selected shared equilibrium is*

$$\frac{7 + 3\rho}{25} \quad \text{in the low shared regime,} \quad \frac{4 + 3\rho}{13 + 12\rho} \quad \text{in the high shared regime,}$$

and is strictly positive.

Centralized top-two is an authority benchmark: it assigns distinct targets. It is not an equilibrium generated by information sharing alone. Opposite constant private equilibria also reach one, but for a different reason and without using the clues. Reporting only the common value would hide the difference between binding action assignment and decentralized strategic coordination.

Communication can also be strategic. Cheap-talk models show that what is sent depends on preferences and incentives [Crawford and Sobel, 1982]; jury and voting models show that information aggregation depends on strategic behavior [Austen-Smith and Banks, 1996, Levy and Razin, 2015]. The current model holds truthful clue sharing fixed and studies the action game that follows. It therefore isolates one implementation boundary but does not solve strategic disclosure.

Boundary. The positive sharing theorem, the selection failure, and the planner gap are jointly necessary for interpretation. Dropping the selection qualifier would change the theorem. Dropping the planner label would conflate information with authority.

9 Design implications and limitations

The results support a diagnostic sequence rather than a universal prescription. First, specify the discovery objective: top-one accuracy, coverage by a portfolio, or expected payoff can rank architectures differently. Second, measure signal geometry through posterior coverage or residual error, not only marginal accuracy. Third, state how many independent actions survive pooling. Fourth, identify who can assign actions and which equilibrium or behavioral rule is being compared. Finally, preserve null and negative findings within their registered scope.

For centralized teams, the action-budget profile is the natural planning object. If $V_1 < P_N$ but $V_2 \geq P_N$, the informational case for sharing may be sound while a one-action implementation is not. The design response could be to reserve a second action, maintain a private red team, or make the pooled output a shortlist rather than a single recommendation. The theorem does not price these options; it reveals where such pricing enters.

For decentralized systems, role design can matter as much as disclosure. Agent identifiers, clue ownership, randomized conventions, and assignment authority change the implementable strategy space. Information sharing can improve one symmetric selection while

an asymmetric or ownership-aware equilibrium performs better. A deployment claim should therefore name its selection mechanism rather than report “the equilibrium.”

Social-learning models explain how repeated observation can create herding, cascades, and overweighting of common sources [Banerjee, 1992, Bikhchandani et al., 1992, DeMarzo et al., 2003]. The current finite models share the concern that correlation and duplication are easy to misread, but they do not estimate a dynamic learning process. Public information can also have coordination benefits or costs in macroeconomic games [Morris and Shin, 2002, Cornand and Heinemann, 2008]; our discovery objective and rescue portfolio produce a different, explicitly action-budgeted frontier.

Information design asks how a designer should choose signals or recommendations [Kamenica and Gentzkow, 2011, Bergemann and Morris, 2016]. We mostly take the registered channels as given. The frontier could become a design constraint—choose a pooled signal whose residual error contracts beyond the rescue threshold—but optimizing over all feasible experiments is outside the verified claims. Team theory likewise asks how information should be allocated under decentralized decision rules [Radner, 1962]. Our contribution is a small exact bridge from pooled coverage to discovery portfolios and selection boundaries.

Several limitations are structural. Rescue actions are independent in the nonstrategic identity; correlated rescue would require their joint failure probability. The finite channel registries do not span arbitrary experiments. The strategic model is binary, two-agent, and uses a specific split-prize payoff. Sharing is truthful and costless. No agent can acquire additional information, and no communication network is modeled. The planner values discovery only and can bind actions. The strategic positive interval fixes $p = 3/5$ and two named equilibrium selections. No welfare claim is made beyond the registered discovery and payoff objects.

The empirical boundary is equally important. No participants were recruited, no human or real organizational data were collected, and no experiment was run. The displayed counts are exact bounded enumerations of synthetic model inputs. They should motivate experimental contrasts, not be described as observations.

The evidence architecture makes extension possible. A new channel family could be registered and classified by its contraction ratios. A richer strategic study could expand the equilibrium correspondence or introduce an explicit selection institution. An empirical study could preregister sharing and action- budget treatments. None of those future directions is silently included here.

10 Conclusion

Information sharing improves decentralized discovery when its aggregation gain is large enough to replace the independent rescue capacity it removes and when the relevant decentralized behavior actually implements that gain. The first condition is captured exactly by the residual-error frontier $e_{s+1}/e_s < 1 - q$. The second requires an equilibrium or institu-

layer	evidence	authority	boundary
Signal geometry	exact bounded	centralized V_L	scalar insufficiency
Incremental sharing	identity/theorem	nonstrategic protocol	arbitrary-channel failure
General frontier	theorem plus census	centralized recovery	mixed-curve bounded null
Strategic sharing	selected BNE theorem	autonomous actions	not every equilibrium
Selection failure	verified negative	broader strategy space	discovery-one alternatives

Figure 8: Evidence and authority map. Analytic, bounded exact, centralized, selected decentralized, and negative-boundary claims remain distinct even when they form one theorem family.

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Runs: 20260722T084145Z_DD-019_a77bb786_04a5e9f0c5, 20260722T142551Z_DD-020_3854fff6_37c11a850a,
20260722T185924Z_DD-021_3cdbbc40_2fea269a9a, 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims:
DD-C-0089-DD-C-0110; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256
prefixes: a7da8a20aa24, 607d03fba773, a4f63d52f402, 0ad5c3249e7d, 3521286a3415, 2bd510abf8cd, e897c62593b4,
98fb7d475365, 8ae5af15a2f7, 146dcfba3f74.

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tional account.

Finite channel examples show why accuracy alone cannot answer the question. Equal one-person accuracy and equal private discovery can coexist with sharply different pooled profiles, recovery budgets, and sharing directions. Full centralized action capacity always recovers the private baseline, but the required budget is part of the design, not a free consequence of communication.

The strategic model supplies a strict positive interval without planner action assignment, but it simultaneously supplies the boundary: the result belongs to named equilibrium selections. Alternative private and ownership-aware strategies and the centralized top-two benchmark prevent an every-equilibrium or pure information interpretation. The useful conclusion is therefore conditional and operational. Share when pooled residual error clears the rescue threshold, retain enough action capacity, and state how diverse actions will be selected.

A Proof details for the sharing identities

This appendix records algebra used by the synthesis; the scientific ownership remains with DD-020 and DD-021. Let F_s denote failure of the pooled block and R_i failure of private rescue action i . Under the registered protocol, $\mathbb{P}(F_s) = 1 - C_s$ and the $N - s$ rescue failures are independent with probability $1 - q$. Therefore

$$\mathbb{P}(F_s \cap R_1 \cap \cdots \cap R_{N-s}) = (1 - C_s)(1 - q)^{N-s}.$$

claim range	owner	evidence class	manuscript role
DD-C-0089-0091	DD-019	independent repository method	geometry and recovery
DD-C-0092-0094	DD-020	identity and verified theorems	aggregation versus rescue
DD-C-0095-0096	DD-020	independent repository method	census and counterchannel
DD-C-0097-0098	DD-021	verified theorems	frontier and centralized recovery
DD-C-0099-0102	DD-021	independent repository method	classes, witnesses, budgets
DD-C-0103	DD-021	verified bounded negative	mixed-curve null
DD-C-0104-0107	DD-022	verified theorem/corollary	selected equilibria and interval
DD-C-0108	DD-022	independent repository method	42-cell classification
DD-C-0109	DD-022	verified negative	selection failure
DD-C-0110	DD-022	verified theorem	implementation gap

Table 9: Claim, evidence, and ownership map. Here independently implemented repository verification is not external replication. Status belongs to each claim, not to the paper as a whole.

Runs: 20260722T084145Z_DD-019_a77bb786_04a5e9f0c5, 20260722T142551Z_DD-020_3854fff6_37c11a850a, 20260722T185924Z_DD-021_3cdbbc40_2fea269a9a, 20260722T210334Z_DD-022_2376d5b7_ad67765ca8; claims: DD-C-0089-DD-C-0110; generator: distributed_discovery.papers.build_information_sharing_frontier; input SHA-256 prefixes: a7da8a20aa24, 607d03fba773, a4f63d52f402, 0ad5c3249e7d, 3521286a3415, 2bd510abf8cd, e897c62593b4, 98fb7d475365, 8ae5af15a2f7, 146dcfba3f74.

Taking one minus this probability proves equation (1). At the next step, one fewer rescue action remains. Subtraction gives

$$\begin{aligned}
G_{s+1} - G_s &= (1 - C_s)(1 - q)^{N-s} - (1 - C_{s+1})(1 - q)^{N-s-1} \\
&= (1 - q)^{N-s-1} [(1 - q)e_s - e_{s+1}].
\end{aligned}$$

The factor outside the brackets is nonnegative and is positive away from the degenerate $q = 1$ boundary. If $e_s > 0$, division by e_s produces the contraction ratio. Notice that $C_{s+1} \geq C_s$ alone is not enough. The absolute gain must be large enough relative to qe_s , the rescue value applied to the current pooled miss probability.

Two boundary cases deserve explicit labels. If $e_s = 0$, the current pooled block already discovers surely; the ratio is undefined but the undivided identity remains valid. If $q = 1$, any remaining rescue action guarantees discovery, so absorbing the last perfect rescuer cannot improve the portfolio. These cases do not affect the registered theorem rows but prevent careless division.

B Canonical exact records

For auditability, the principal frozen profiles are restated here. DD-019 records

$$\begin{array}{lll}
\text{noisy point: } (7/12, 43/54, 25/27), & P_3 = 7/8, & L^* = 3, \\
\text{noisy shortlist: } (35/64, 79/96, 539/576), & P_3 = 387/512, & L^* = 2, \\
\text{guaranteed shortlist: } (17/18, 1, 1), & P_3 = 7/8, & L^* = 1, \\
\text{exclusion: } (16/27, 26/27, 1), & P_3 = 19/27, & L^* = 2, \\
\text{confidence point: } (295/384, 175/192, 31/32), & P_3 = 485/512, & L^* = 3.
\end{array}$$

The associated DD-020 paths are respectively

$$\begin{aligned} & (7/8, 3/4, 7/12), \quad (387/512, 11/16, 35/64), \\ & (7/8, 11/12, 17/18), \quad (19/27, 17/27, 16/27), \\ & (485/512, 113/128, 295/384). \end{aligned}$$

These are exact source values, not numbers reconstructed from the plotted lines.

At $p = 3/5$, DD-022 records the selected discovery comparisons at $\rho \in \{0, 1/6, 1/4, 1/2, 7/12, 3/4, 1\}$. The private selected sequence is $21/25, 4/5, 39/50, 18/25, 7/10, 27/38, 18/25$; the shared selected sequence is $18/25, 7/10, 45/64, 27/38, 57/80, 63/88, 18/25$. The source CSV distributed with figure 6 includes these cells and the centralized value one.

C Protocol-level derivations and interpretation checks

This appendix makes explicit the sequence of objects that can otherwise be compressed into the phrase “sharing improves discovery.” It introduces no new claim; it is an editorial derivation and interpretation audit of DD-019 through DD-022.

From a channel to an action-budget profile

Fix a prior and a finite signal history h . Bayes’ rule produces posterior masses $\pi_j(h)$ over candidates. Sort them as $\pi_{(1)}(h) \geq \dots \geq \pi_{(M)}(h)$. Conditional on h , a centralized planner with L distinct actions covers the correct candidate with probability $\sum_{j=1}^L \pi_{(j)}(h)$. Averaging over histories produces V_L . This calculation uses the complete posterior vector. Keeping only $\max_j \pi_j(h)$ would determine V_1 but discard the marginal value of actions two through M .

That lost information explains the equal-accuracy example. Accuracy constrains the expected success of a registered top action. It does not determine whether the posterior error is concentrated on one runner-up or spread over several candidates. A guaranteed shortlist can put essentially all mass into a known small set even when its top-label correctness is only one half. A noisy point can leave substantial mass outside the first two ranks. Because discovery is a coverage objective, the ordering of the remaining masses matters directly.

The profile is centralized by construction. A decentralized team could attain the same set of actions if it had an implementation device: binding assignments, public roles, or a supported equilibrium that maps agents to distinct ranks. Absent such a device, V_L remains a benchmark. This label prevents a common category error in which a planner’s ordered posterior sum is reported as the automatic consequence of agents observing the same evidence.

From a profile to an incremental-sharing path

The registered incremental protocol chooses a block size s . Its pooled block has a specified coverage C_s ; the remaining agents act as independent rescue attempts. The comparison between s and $s + 1$ therefore changes both the pooled history and the number of rescue attempts. Holding the latter fixed would answer a different question: the marginal value of an additional signal with no capacity cost. Holding the pooled coverage fixed would answer the marginal value of an independent attempt. The sharing frontier is the net comparison.

The failure representation is useful because discovery is a union event. The current architecture fails only if the pooled block fails and every rescue attempt fails. The next architecture may have a smaller pooled failure probability, but it also contains one fewer multiplicative rescue factor. The break-even equation

$$e_{s+1} = (1 - q)e_s$$

says that the pooled block must remove the same share of its residual error as a private attempt would have removed. A strict inequality determines the sign without requiring the absolute level of G_s .

The ratio formulation is scale-free but not assumption-free. If rescue attempts are correlated, $(1 - q)^{N-s}$ must be replaced by their joint conditional failure probability. If rescue quality depends on the pooled history, the common q must be indexed by that history. If pooled agents retain individual actions, the capacity transition is no longer one-for-one. Each extension is feasible, but none is silently covered by DD-C-0097.

From a theorem to a bounded classification

The analytic frontier gives a predicate for each applicable step. DD-021 then applies that predicate to a versioned registry. A curve is called aggregation dominated when each strict step lies below the rescue threshold, compression dominated when each strict step lies above it, all-neutral when equality holds throughout, and mixed when different steps have different strict signs. The class labels are summaries of exact records, not primitives in the theorem.

The distinction between proof and census is important. A test that recomputes all 177 rows can detect implementation drift and show that the theorem agrees with those fixtures. It cannot prove the formula for unenumerated parameters. Conversely, the algebraic proof does not imply that the registry contains a mixed witness. Keeping the analytic and bounded layers separate is what allows the zero mixed count to remain a useful negative result without becoming an unsupported universal monotonicity claim.

Minimal witnesses are also registry-relative. The word “minimal” refers to the frozen lexicographic ordering over registered parameters. It does not mean minimal among every mathematically possible channel or in description length. This qualifier is carried into table 5 and the generated asset note.

From private clues to selected equilibria

The DD-022 comparison begins with a hidden mixture over source dependence. The parameter ρ is the prior probability of the common-source branch. Observing two agreeing clues raises confidence in their label and changes beliefs about which branch was likely, but it does not reveal the branch realization. A strategy that conditions on “the clues were common” would therefore use information unavailable to the agents.

Before sharing, each agent sees one clue and not the other’s. Anonymous label-equivariance requires the same response after relabeling targets. The selected symmetric equilibrium can then be summarized by a clue-following probability r . The statistic $A = t^2 + \rho(1 - t^2)$ captures the unconditional alignment created by clue accuracy and source dependence. At $A = 3t$, the boundary solution gives way to interior anti-crowding. The exact piecewise formula in table 6 is the registered selection.

After sharing, the information histories are agreement on a label or disagreement. At agreement, u is the posterior probability that the agreed label is correct. Identical posterior-only mixing is summarized by x . At disagreement the posterior over targets is one half, so the selected identical mix is one half. This last step is where ownership information is deliberately discarded: although the common posterior is symmetric, agent one and agent two can still know which private clue each originally owned.

The positive interval compares the discovery induced by these two selections. It is meaningful because both are exact Bayesian Nash equilibria under their registered restrictions. It is limited because equilibrium existence does not select them from all alternatives. An opposite constant-target profile ignores the signals but covers the binary state space. An ownership-aware rule can use agent identity to split actions after disagreement. These constructions do not invalidate the selected interval; they invalidate a stronger interpretation of it.

Three meanings of discovery one

The value one appears in three places that must not be collapsed. First, a centralized planner with two actions covers both binary targets by binding assignment. Second, opposite constant-target private equilibria cover both targets through a fixed decentralized convention. Third, an ownership-aware shared rule can cover both targets on disagreement histories while behaving differently on agreement. The same discovery number masks different evidence use, authority, and off-path behavior.

For organizational design, this means that performance tables need an authority column. A benchmark labeled “centralized top-two” is evidence about attainable coverage under control, not about spontaneous coordination. A row labeled “selected shared equilibrium” is evidence about one decentralized behavioral rule. A row labeled “alternative equilibrium” is evidence about multiplicity. The generated selection map and evidence-authority map enforce those labels visually.

Reading the canonical plot

In figure 6, the dashed direct-private curve is not the private selected equilibrium after the private anti-crowding boundary. The selected private curve initially follows direct clue use, then changes when mixing becomes optimal. The selected shared curve changes formula at its own boundary $1/6$. Its intersection with the private selected curve at ρ^* is algebraic, not a linear interpolation of plotted cells. The second equality at $\rho = 1$ reflects the fact that perfectly common clues contain no additional cross-agent evidence when shared.

The connected points in the figure are a display of registered exact cells. The theorem between them comes from piecewise symbolic formulas and a certified root isolation. The decimal marker is only a visual aid. The exact expression and rational bracket control every sign claim.

Interpretation checklist

For each proposed use of the results, the following questions identify the relevant row of the evidence map:

1. Is the outcome top-choice accuracy or coverage by at least one action?
2. Does sharing reduce the number of independent actions, and by how much?
3. Are remaining rescue attempts independent conditional on pooled failure?
4. Is the displayed action budget feasible and centrally assignable?
5. If actions are decentralized, which equilibrium selection or role system maps shared evidence to distinct actions?
6. Is a reported count an analytic theorem, a bounded exact census, or a preserved null inside a registered search space?
7. Does the protocol reveal only the pooled clues, or also an unavailable latent source branch?
8. Are claims about model-generated values kept separate from claims about people, organizations, or deployed multi-agent systems?

An application that cannot answer these questions has not yet specified the comparison made by this paper.

D Evidence, provenance, and reproducibility

The generator resolves all 22 claim IDs against the claim ledger, verifies the ten source-output hashes against immutable run manifests, writes exact CSV data for every figure, and writes TeX tables and figures with run IDs, claim IDs, the generator name, and input checksum prefixes. It then builds the PDF twice under a fixed source epoch and requires byte-identical hashes. An independent paper audit rechecks the source contracts without importing the generator and performs deliberate corruption tests on copied metadata.

Here “independent” means independently implemented repository verification: the second method or audit does not import or call the primary generator path. It is not external replication by an unaffiliated team. The complete immutable records, independent verifier outputs, deliberate corruption checks, and paper generator are public at repository commit 66ba4449572b20c519a4629ef20cfc030fae1762. The four run directories and their complete manifest SHA-256 digests are:

- https://github.com/yoheinakajima/distributed-discovery/tree/66ba4449572b20c519a4629ef20cfc030fae1762/results/verified/20260722T084145Z_DD-019_a77bb786_04a5e9f0c5,011c7699e6bf2bb90b1201797a430b88b74228385e74df61d23d03a124670c53;
- https://github.com/yoheinakajima/distributed-discovery/tree/66ba4449572b20c519a4629ef20cfc030fae1762/results/verified/20260722T142551Z_DD-020_3854fff6_37c11a850a,82f8c71d65b27a48b075d06d0a186040cc9e6015aaee9bc695ff2e9fa823a972;
- https://github.com/yoheinakajima/distributed-discovery/tree/66ba4449572b20c519a4629ef20cfc030fae1762/results/verified/20260722T185924Z_DD-021_3cdbbc40_2fea269a9a,69ac4c760d7a34b6ee315ae619287b7928f69cc8a20eaf4a62202868baadcacd;
- https://github.com/yoheinakajima/distributed-discovery/tree/66ba4449572b20c519a4629ef20cfc030fae1762/results/verified/20260722T210334Z_DD-022_2376d5b7_ad67765ca8,3d905723eda6069cf870f1fb7b2a241cfa592ea43e9009473bd2b0878bdd5bb1.

The principal review-facing outputs are DD-020 `outputs/channel-profiles.json` at a4f63d52f402ce14f91e469d4b425ce65a92740b31530c7da9384cc726030e3a, DD-021 `outputs/registry.json` at 3521286a341595d27b95248a0720f4df672bb1b811d17edf7f286b5520912697, and DD-022 `outputs/registry.json` at 98fb7d475365548fcc723603b3c3721e673b0b0423bc89579c411c3252ac54f1. Their run directories also contain the full verification and corruption-test records. The paper generator is https://github.com/yoheinakajima/distributed-discovery/blob/66ba4449572b20c519a4629ef20cfc030fae1762/src/distributed_discovery/papers/build_information_sharing_frontier.py; the arXiv source package includes compact copies of the cited immutable evidence.

The manuscript makes no new claim record and no new study or run. DD-019 owns the signal-geometry results, DD-020 the incremental identity and counterchannel, DD-021 the general frontier and recovery registry, and DD-022 the strategic equilibrium results. Editorial synthesis does not change those ownership lines. The citation database is copied into the generated bundle at build time so the compiled bibliography is pinned to the paper artifact.

E Literature boundary

The adjacent literatures establish why narrow contribution language is needed. Blackwell comparison concerns informativeness across decision problems [Blackwell, 1953]. Team decision theory concerns decentralized information allocation [Radner, 1962]. Herding and social-learning models concern the endogenous informational content of observed behavior [Banerjee, 1992, Bikhchandani et al., 1992, Acemoglu et al., 2011]. Strategic communication concerns incentive-compatible messages [Crawford and Sobel, 1982]. Information design concerns signal or recommendation choice [Kamenica and Gentzkow, 2011, Bergemann and Morris, 2016]. Equilibrium selection concerns which equilibrium is predicted when several exist [Harsanyi and Selten, 1988].

The present synthesis composes a particular portfolio objective, rescue technology, coverage profile, and selected action game. That composition may be useful without supporting a claim that no related frontier, anti-crowding mechanism, or selection caveat exists elsewhere. The accompanying literature files record stable identifiers and the claim-by-claim relationship.

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